

Diffusive Stochastic Master Equation (SME) with dispersive qubit/cavity coupling

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Abstract A detailed analysis of the quantum diffusive Stochastic Master Equation (SME) for qubit/cavity systems with dispersive coupling is provided. This analysis incorporates classical input signals and output signals (measurement outcomes through homodyne/heterodyne detection). The dynamics of the qubit/cavity density operator is shown to converge exponentially towards a slow invariant manifold. This invariant manifold is parameterized by the density operator of a fictitious qubit governed by a quantum SME incorporating the classical input/output signals preserving complete-positivity and trace. The reduced cavity (resp. qubit) operator obtained by partial trace of the qubit (resp. cavity) is then given by a time-varying deterministic quantum channel from the density operator of this fictitious qubit. This formulation avoids non-Markovian descriptions where negative dephasing rates and detection efficiency exceeding temporary one are present in several publications. Extensions are considered where the qubit is replaced by any qudit dispersively coupled to an arbitrary set of cavity modes with collective input/output classical signals.

1 Introduction

Quantum non-demolition measurements [14, 8] play a key role in quantum error correction. Syndrome error detections are based on joint parity measurements [10]. For superconducting qubits, they rely on homodyne/heterodyne setups (see, e.g., [7, 1]) governed by quantum SME Stochastic Master Equations (SME, for a mathematical exposure see [2] and tutorial ones see, e.g., [6, 12]). These models share a common structure that has been explored initially in [5], where a polaron transformation (a time-varying unitary change of frame) is used. The ensemble-average evolution of the density operator for the qubit is shown to obey a kind of Lindblad master equation but where the dephasing rate could

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be temporary negative ($\gamma_\phi + \Gamma_d(t)$ in [5, equation (3.10)]). In the physics community this indicates non-Markovian dynamics. Several publications follow the seminal publication [5] with different setup configurations [9, 4, 13]. In particular extensions to qudit with several cavity modes are investigated. These extensions exploit generalized polaron transformations but the non-Markovian property of qudit dynamics remains (see, e.g., [15, equations (47,48,49)]).

We propose here a slightly different formulation for the evolution of the qudit density operator. It is based on a fully Markovian evolution (either deterministic or stochastic) of a fictitious qudit from which the qudit operator is derived via a time-varying quantum channel (see equation (6) for qubit/cavity). Such formulation avoids possible negativity in dephasing rates and detection efficiency possibly greater than one. It is based on a invariant manifold (lemma 1 for qubit/cavity) which is shown to be exponentially attractive with a rate directly related to the measurement strength (lemma 2 for qubit/cavity). Such manifold provides an exact reduced-order model, simplifies the analysis of such measurement setup and preserves the complete-positivity of the qudit evolution. From a control theory view point, this quantum channel can be seen as an output map: (6) provides ρ_Q , the qubit operator seen here as an output, form ξ_Q , the fictitious qubit operator seen here as the state governed by the stochastic Markovian dynamics (5). Such point of view bridges quantum dynamics with control theory concepts, facilitating the design of interfaces between quantum measurement with control analysis and design. By formalizing the connection between quantum stochastic dynamics and usual control paradigms, this work opens new avenues for the development of high-performance quantum control systems, with applications ranging from quantum error correction to precision metrology.

The structure of the paper is as follows. Section 2 focus on the qubit/cavity system considered originally in [5] where dephasing rate could be negative. Section 3 deals with the qudit extension considered in [4]. Last section 4 shows how to treat a qudit with several cavity modes as in [4, 15]. We focus in this paper on homodyne detection. Similar results hold also true for heterodyne detection.

2 The qubit/cavity case

Following [5], one starts with the following SME for the density operator ρ of the qubit/cavity system

$$d\rho = \left(-i \left[\sqrt{\kappa}u\mathbf{a}^\dagger + \sqrt{\kappa}u^*\mathbf{a} + \chi\boldsymbol{\sigma}_z\mathbf{a}^\dagger\mathbf{a}, \rho\right] + \kappa\mathcal{D}_\mathbf{a}(\rho)\right) dt + \sqrt{\eta\kappa}\mathcal{M}_\mathbf{a}(\rho) dw \quad (1)$$

where the classical input signal is $u(t) \in \mathbb{C}$ (cavity coherent drive) and the classical measurement outcome signal y obeys to $dy_t = \sqrt{\eta\kappa}\text{Tr}(\mathbf{a}\rho + \rho\mathbf{a}^\dagger)dt + dw_t$. Here $\mathbf{a}$ is the photon annihilation operator, $\boldsymbol{\sigma}_z$ the Pauli operator associated to the qubit energy, and dw is a classical Wiener process, $\mathcal{D}_\mathbf{L}(\rho) \equiv \mathbf{L}\rho\mathbf{L}^\dagger - (\mathbf{L}^\dagger\mathbf{L}\rho + \rho\mathbf{L}^\dagger\mathbf{L})/2$ and $\mathcal{M}_\mathbf{L}(\rho) \equiv \mathbf{L}\rho + \rho\mathbf{L}^\dagger - \text{Tr}(\mathbf{L}\rho + \rho\mathbf{L}^\dagger)\rho$. The real parameters are the dispersive coupling strength χ , the measurement rate $\kappa > 0$ and the detection efficiency $\eta \in [0, 1]$.

As in [5], we consider the following change of frame

$$\rho = (\mathbf{P}_g \mathbf{D}_{\alpha_g} + \mathbf{P}_e \mathbf{D}_{\alpha_e}) \boldsymbol{\xi} (\mathbf{P}_g \mathbf{D}_{-\alpha_g} + \mathbf{P}_e \mathbf{D}_{-\alpha_e})$$

with α_g, α_e being time-varying complex signals, $\mathbf{P}_g = |g\rangle\langle g|$, $\mathbf{P}_e = |e\rangle\langle e|$. The unitary displacement of complex amplitude α corresponds to $\mathbf{D}_\alpha = e^{\alpha \mathbf{a}^\dagger - \alpha^* \mathbf{a}}$. With

$$\frac{d}{dt} \alpha_g = i\chi \alpha_g - i\sqrt{\kappa}u - \frac{\kappa}{2} \alpha_g \quad (2)$$

$$\frac{d}{dt} \alpha_e = -i\chi \alpha_e - i\sqrt{\kappa}u - \frac{\kappa}{2} \alpha_e \quad (3)$$

one gets after some simple but tedious calculations

$$\begin{aligned} d\boldsymbol{\xi} = & -i \left[\mathbf{P}_g \left(\Re(\alpha_g \sqrt{\kappa}u^*) - \chi \mathbf{a}^\dagger \mathbf{a} + \frac{i\kappa}{2} (\alpha_g \mathbf{a}^\dagger - \alpha_g^* \mathbf{a}) \right), \boldsymbol{\xi} \right] dt \\ & - i \left[\mathbf{P}_e \left(\Re(\alpha_e \sqrt{\kappa}u^*) + \chi \mathbf{a}^\dagger \mathbf{a} + \frac{i\kappa}{2} (\alpha_e \mathbf{a}^\dagger - \alpha_e^* \mathbf{a}) \right), \boldsymbol{\xi} \right] dt \\ & + \kappa \mathcal{D}_{\alpha_g \mathbf{P}_g + \alpha_e \mathbf{P}_e}(\boldsymbol{\xi}) dt + \sqrt{\eta\kappa} \mathcal{M}_{\alpha_g \mathbf{P}_g + \alpha_e \mathbf{P}_e}(\boldsymbol{\xi}) dw \end{aligned} \quad (4)$$

with $dy = \sqrt{\eta\kappa} \text{Tr}((\alpha_g \mathbf{P}_g + \alpha_e \mathbf{P}_e) \boldsymbol{\xi} + \boldsymbol{\xi}(\alpha_g^* \mathbf{P}_g + \alpha_e^* \mathbf{P}_e)) dt + dw$.

We have the following lemma underlying the exact reduced-order model introduced in [5].

Lemma 1. *Assume the following initial condition $\boldsymbol{\xi}(0) = \boldsymbol{\xi}_Q^0 \otimes |0\rangle\langle 0|$. Then for any time $t > 0$, $\boldsymbol{\xi}(t) = \boldsymbol{\xi}_Q(t) \otimes |0\rangle\langle 0|$ with $\boldsymbol{\xi}_Q$ being governed by the following standard SME*

$$\begin{aligned} d\boldsymbol{\xi}_Q = & -i \left[\Re(\alpha_g \sqrt{\kappa}u^*) \mathbf{P}_g + \Re(\alpha_e \sqrt{\kappa}u^*) \mathbf{P}_e, \boldsymbol{\xi}_Q \right] dt + \kappa \mathcal{D}_{\alpha_g \mathbf{P}_g + \alpha_e \mathbf{P}_e}(\boldsymbol{\xi}_Q) dt \\ & + \sqrt{\eta\kappa} \mathcal{M}_{\alpha_g \mathbf{P}_g + \alpha_e \mathbf{P}_e}(\boldsymbol{\xi}_Q) dw \end{aligned} \quad (5)$$

with output $dy = \sqrt{\eta\kappa} \text{Tr}((\alpha_g \mathbf{P}_g + \alpha_e \mathbf{P}_e) \boldsymbol{\xi}_Q + \boldsymbol{\xi}_Q(\alpha_g^* \mathbf{P}_g + \alpha_e^* \mathbf{P}_e)) dt + dw$ and starting from $\boldsymbol{\xi}_Q(0) = \boldsymbol{\xi}_Q^0$.

This lemma means that the set $\mathcal{I}_Q = \{\boldsymbol{\xi}_Q \otimes |0\rangle\langle 0|\}$ where $\boldsymbol{\xi}_Q$ is any qubit density operator, is invariant for any realisation of (4). We have this second lemma showing the convergence of (4) towards $\mathcal{I}_Q = \{\boldsymbol{\xi} \mid \text{Tr}(\mathbf{a}^\dagger \mathbf{a} \boldsymbol{\xi}) = 0\}$.

Lemma 2. *Any solution of (4) converges exponentially and almost surely towards the invariant set $\mathcal{I}_Q$ since the super-martingale $\text{Tr}(\mathbf{a}^\dagger \mathbf{a} \boldsymbol{\xi})$ satisfies*

$$\frac{d}{dt} \mathbb{E}(\text{Tr}(\mathbf{a}^\dagger \mathbf{a} \boldsymbol{\xi})) = -\kappa \mathbb{E}(\text{Tr}(\mathbf{a}^\dagger \mathbf{a} \boldsymbol{\xi})).$$

The proof is based on standard but formal computations avoiding infinite dimensional technicalities. A rigorous mathematical justification will be based on functional analysis to ensure wellposedness and exponential convergence with adapted norms. Such justifications can certainly be done with the frame-work developed in [3].

Thus to describe the classical input u and classical output y dynamics when the initial qubit/cavity is $\rho(0) = \rho_Q^0 \otimes |\alpha^0\rangle\langle\alpha^0|$ ($\alpha^0 \in \mathbb{C}$ arbitrary), we have a state space representation (analogue of a Kalman input/output representation). It is based on (2) and (3) with the common initial condition α^0 and on (5) with initial condition $\boldsymbol{\xi}_Q(0) = \rho_Q^0$. The qubit/cavity density operator ρ at time t is then given by the above unitary transformation, seen here as an output map depending only on the state $(\alpha_g, \alpha_e, \boldsymbol{\xi}_Q)$ at time t :

$$\rho(t) = (\mathbf{P}_g \mathbf{D}_{\alpha_g(t)} + \mathbf{P}_e \mathbf{D}_{\alpha_e(t)}) (\boldsymbol{\xi}_Q(t) \otimes |0\rangle\langle 0|) (\mathbf{P}_g \mathbf{D}_{-\alpha_g(t)} + \mathbf{P}_e \mathbf{D}_{-\alpha_e(t)}).$$

A partial trace versus the qubit gives the non-Markovian evolution of $\rho_C = \text{Tr}_{\text{qubit}}(\rho)$ via the following cavity output map

$$\rho_C(t) = \text{Tr}(\mathbf{P}_g \boldsymbol{\xi}_Q(t)) |\alpha_g(t)\rangle\langle\alpha_g(t)| + \text{Tr}(\mathbf{P}_e \boldsymbol{\xi}_Q(t)) |\alpha_e(t)\rangle\langle\alpha_e(t)|$$

A partial trace versus the cavity gives the non-Markovian evolution of $\rho_Q = \text{Tr}_{\text{cavity}}(\rho)$ via the following qubit output map

$$\rho_Q(t) = (\mathbf{P}_g + c_{ge}(t)\mathbf{P}_e)\boldsymbol{\xi}_Q(t)(\mathbf{P}_g + c_{ge}^*(t)\mathbf{P}_e) + (1 - |c_{ge}(t)|^2)\mathbf{P}_e\boldsymbol{\xi}_Q(t)\mathbf{P}_e \quad (6)$$

where $c_{ge} = \langle\alpha_g|\alpha_e\rangle = e^{-\frac{|\alpha_g|^2 + |\alpha_e|^2}{2} + \alpha_g^* \alpha_e}$. Notice that this qubit output map is just a Kraus map from $\boldsymbol{\xi}_Q$ to ρ_Q with two time-varying Kraus operators $\mathbf{P}_g + c_{ge}(t)\mathbf{P}_e$ and $\sqrt{1 - |c_{ge}(t)|^2}\mathbf{P}_e$.

Such a formulation of the dynamics avoids for ρ_Q a SME description with detection efficiencies greater than one during some time intervals. Here $\boldsymbol{\xi}_Q$ obeys to a standard stochastic master equation with a constant detection efficiency $\eta \in [0, 1]$.

3 Qudit extension

This extension is inspired from [4]. It can be seen as a state-space reformulation combining usual diffusive SME with constant detection efficiency, linear deterministic differential equations and Kraus maps to represent the non-Markovian evolution of the density operator attached to each subsystem.

The above developments generalize when the qubit Q is replaced by a finite dimensional system $\mathcal{S}$ of Hilbert basis $(|s\rangle)_{s=1,\dots,n_{\mathcal{S}}}$ with

$$d\rho = \left(-i \left[\sqrt{\kappa} u \mathbf{a}^\dagger + \sqrt{\kappa} u^* \mathbf{a} + \left(\sum_s \chi_s \mathbf{P}_s \right) \mathbf{a}^\dagger \mathbf{a}, \rho \right] + \kappa \mathcal{D}_{\mathbf{a}}(\rho) \right) dt + \sqrt{\eta \kappa} \mathcal{M}_{\mathbf{a}}(\rho) dw$$

where $\mathbf{P}_s = |s\rangle\langle s|$ and χ_s real parameter. With coherent amplitudes α_s given by

$$\frac{d}{dt}\alpha_s = -i\chi_s\alpha_s - i\sqrt{\kappa}u - \frac{\kappa}{2}\alpha_s$$

and unitary transformation

$$\rho = \left(\sum_s P_s D_{\alpha_s} \right) \xi \left(\sum_s P_s D_{-\alpha_s} \right)$$

one gets

$$d\xi = -i \left[\sum_s P_s \left(\Re(\alpha_s \sqrt{\kappa} u^*) + \chi_s \mathbf{a}^\dagger \mathbf{a} + \frac{i\kappa}{2} (\alpha_s \mathbf{a}^\dagger - \alpha_s^* \mathbf{a}) \right), \xi \right] dt \\ + \kappa \mathcal{D}_{\mathbf{a} + \sum_s \alpha_s P_s}(\xi) dt + \sqrt{\eta\kappa} \mathcal{M}_{\mathbf{a} + \sum_s \alpha_s P_s}(\xi) dw \quad (7)$$

with $dy = \sqrt{\eta\kappa} \text{Tr} \left((\mathbf{a} + \sum_s \alpha_s P_s) \xi + \xi (\mathbf{a}^\dagger + \sum_s \alpha_s^* P_s) \right) dt + dw$.

Then lemma 1 extends to

Lemma 3. Assume the following initial condition $\xi(0) = \xi_S^0 \otimes |0\rangle\langle 0|$. Then for any time $t > 0$, $\xi(t) = \xi_S(t) \otimes |0\rangle\langle 0|$ with ξ_S being governed by the following standard SME

$$d\xi_S = -i \left[\sum_s \Re(\alpha_s \sqrt{\kappa} u^*) P_s, \xi_S \right] dt + \kappa \mathcal{D}_{\sum_s \alpha_s P_s}(\xi_S) dt + \sqrt{\eta\kappa} \mathcal{M}_{\sum_s \alpha_s P_s}(\xi_S) dw$$

with output $dy = \sqrt{\eta\kappa} \text{Tr} \left((\sum_s \alpha_s P_s) \xi_S + \xi_S (\sum_s \alpha_s^* P_s) \right) dt + dw$ and $\xi_S(0) = \xi_S^0$.

We have this second lemma showing the convergence of (7) towards $\mathcal{I}_S = \{\xi \mid \text{Tr}(\mathbf{a}^\dagger \mathbf{a} \xi) = 0\}$.

Lemma 4. Any solution of (7) converges exponentially and almost surely towards the invariant set $\mathcal{I}_S = \{\xi_S \otimes |0\rangle\langle 0|\}$ where ξ_S is any density operator on $\mathcal{S}$. More precisely we have

$$\frac{d}{dt} \mathbb{E}(\text{Tr}(\mathbf{a}^\dagger \mathbf{a} \xi)) = -\kappa \mathbb{E}(\text{Tr}(\mathbf{a}^\dagger \mathbf{a} \xi)).$$

The state space input/output description when the initial system/cavity is $\rho(0) = \rho_S^0 \otimes |\alpha^0\rangle\langle \alpha^0|$ ($\alpha^0 \in \mathbb{C}$ arbitrary) is given by ξ_S with initial condition ρ_S^0 and by $(\alpha_s)_{s=1, \dots, n_S}$ with the common initial condition α^0 . The partial trace versus the system gives the non-Markovian evolution of $\rho_C = \text{Tr}_{\text{system}}(\rho)$ via the following cavity output map

$$\rho_C(t) = \sum_s \text{Tr}(P_s \xi_S(t)) |\alpha_s(t)\rangle\langle \alpha_s(t)|$$

The partial trace versus the cavity gives the non-Markovian evolution of $\rho_S = \text{Tr}_{\text{cavity}}(\rho)$ via the following output map

$$\rho_S(t) = \sum_{s, s'} c_{s's}(t) P_s \xi_S(t) P_{s'}$$

where $c_{s's} = \langle \alpha_{s'} | \alpha_s \rangle$. Since the square matrix $C = (c_{s's})_{1 \leq s', s \leq n_C}$ is the Gram matrix of the n_C states $|\alpha_s\rangle$, it is Hermitian and non negative. Denote by $r_{s', s}$ the entries of its

square root $\sqrt{C} = (r_{s',s})_{1 \leq s',s \leq n_C}$. One gets then a usual Kraus map formulation with n_C operators $\mathbf{K}_s$:

$$\rho_S(t) = \sum_s \mathbf{K}_s(t) \xi_S(t) \mathbf{K}_s^\dagger(t) \quad \text{where} \quad \mathbf{K}_s(t) = \sum_{s'} r_{s,s'}(t) \mathbf{P}_{s'}.$$

Notice that, by construction $r_{s,s'}^* \equiv r_{s',s}$ and $\sum_{s'} r_{s,s'} r_{s',s} = \sum_{s'} |r_{s,s'}|^2 = 1$.

4 Multi-cavity extension

This extension is inspired from [4, 15]. The above developments generalize when cavity mode $\mathbf{a}$ is replaced by n_C modes labelled via $c = 1, \dots, n_C$ with annihilation operator $\mathbf{a}_c$. The system $\mathcal{S}$ lives in the Hilbert space of basis $(|s\rangle)_{s=1, \dots, n_S}$. With dispersive coupling between $\mathcal{S}$ and these n_C modes, with the collective coherent input u and with the collective homodyne detection operator $\mathbf{a}_{out} = \sum_c \sqrt{\kappa_c} \mathbf{a}_c$, one starts from the following SME,

$$d\rho = \left(-i \left[\sum_c \left(\sqrt{\kappa_c} u \mathbf{a}_c^\dagger + \sqrt{\kappa_c} u^* \mathbf{a}_c + \left(\Delta_c + \sum_s \chi_{s,c} \mathbf{P}_s \right) \mathbf{a}_c^\dagger \mathbf{a}_c \right), \rho \right] \right) dt + \mathcal{D}_{\sum_c \sqrt{\kappa_c} \mathbf{a}_c}(\rho) dt + \sqrt{\eta} \mathcal{M}_{\sum_c \sqrt{\kappa_c} \mathbf{a}_c}(\rho) dw$$

where $\mathbf{P}_s = |s\rangle\langle s|$ and $\kappa_c > 0$, Δ_c , $\chi_{s,c}$, $\eta \in [0, 1]$ real parameters. Consider coherent amplitudes $\alpha_{s,c}$ given by

$$\frac{d}{dt} \alpha_{s,c} = -i(\Delta_c + \chi_{s,c}) \alpha_{s,c} - i\sqrt{\kappa_c} u - \sum_{c'} \frac{\sqrt{\kappa_c \kappa_{c'}}}{2} \alpha_{s,c'} \quad (8)$$

and the unitary transformation

$$\rho = \left(\sum_s \mathbf{P}_s \mathbf{D}_{\alpha_s} \right) \xi \left(\sum_s \mathbf{P}_s \mathbf{D}_{-\alpha_s} \right)$$

where $\alpha_s = (\alpha_{s,1}, \dots, \alpha_{s,n_C})$ and $\mathbf{D}_{\alpha_s} = \exp(\sum_c \alpha_{s,c} \mathbf{a}_c^\dagger - \alpha_{s,c}^* \mathbf{a}_c)$.

With

$$\sqrt{\bar{\kappa}} \bar{\alpha}_s = \sum_c \sqrt{\kappa_c} \alpha_{s,c} \quad \text{and} \quad \sqrt{\bar{\kappa}} = \sum_c \sqrt{\kappa_c}$$

one gets

$$d\xi = -i \left[\sum_s \mathbf{P}_s \left(\Re(\sqrt{\bar{\kappa}} \bar{\alpha}_s u^*) + \sum_c (\Delta_c + \chi_{s,c}) \mathbf{a}_c^\dagger \mathbf{a}_c + \frac{i\sqrt{\bar{\kappa} \kappa_c}}{2} (\bar{\alpha}_s \mathbf{a}_c^\dagger - \bar{\alpha}_s^* \mathbf{a}_c) \right), \xi \right] dt + \mathcal{D}_{\sum_s \mathbf{P}_s (\sqrt{\bar{\kappa}} \bar{\alpha}_s + \sum_c \sqrt{\kappa_c} \mathbf{a}_c)}(\xi) dt + \sqrt{\eta} \mathcal{M}_{\sum_s \mathbf{P}_s (\sqrt{\bar{\kappa}} \bar{\alpha}_s + \sum_c \sqrt{\kappa_c} \mathbf{a}_c)}(\xi) dw \quad (9)$$

with

$$dy = \sqrt{\eta} \text{Tr} \left(\left(\sum_s \mathbf{P}_s \left(\sqrt{\bar{\kappa}} \bar{\alpha}_s + \sum_c \sqrt{\kappa_c} \mathbf{a}_c \right) \right) \boldsymbol{\xi} + \boldsymbol{\xi} \left(\sum_s \mathbf{P}_s \left(\sqrt{\bar{\kappa}} \bar{\alpha}_s^* + \sum_c \sqrt{\kappa_c} \mathbf{a}_c^\dagger \right) \right) \right) dt + dw.$$

Then lemma 3 extends to

Lemma 5. *Assume the following initial condition $\boldsymbol{\xi}(0) = \boldsymbol{\xi}_S^0 \otimes (\otimes_c |0\rangle\langle 0|)$. Then for any time $t > 0$, $\boldsymbol{\xi}(t) = \boldsymbol{\xi}_S(t) \otimes (\otimes_c |0\rangle\langle 0|)$ with $\boldsymbol{\xi}_S$ being governed by the following standard SME*

$$d\boldsymbol{\xi}_S = -i \left[\sum_s \Re(\sqrt{\bar{\kappa}} \bar{\alpha}_s u^*) \mathbf{P}_s, \boldsymbol{\xi}_S \right] dt + \bar{\kappa} \mathcal{D}_{\sum_s \bar{\alpha}_s \mathbf{P}_s}(\boldsymbol{\xi}_S) dt + \sqrt{\eta \bar{\kappa}} \mathcal{M}_{\sum_s \bar{\alpha}_s \mathbf{P}_s}(\boldsymbol{\xi}_S) dw$$

with output $dy = \sqrt{\eta \bar{\kappa}} \text{Tr}((\sum_s \bar{\alpha}_s \mathbf{P}_s) \boldsymbol{\xi}_S + \boldsymbol{\xi}_S (\sum_s \bar{\alpha}_s^* \mathbf{P}_s)) dt + dw$ and $\boldsymbol{\xi}_S(0) = \boldsymbol{\xi}_S^0$.

The convergence is a little more tricky and related to the convergence of n_S systems indexed by s and made of the n_C coupled equations (8) indexed by c . However, we have the following simple stability lemma

Lemma 6. *Any solution of (9) tends to converge almost surely towards the invariant set $\mathcal{I}_S = \{\boldsymbol{\xi}_S \otimes (\otimes_c |0\rangle\langle 0|)\}$ where $\boldsymbol{\xi}_S$ is any density operator on $\mathcal{S}$:*

$$\frac{d}{dt} \left(\sum_c \mathbb{E}(\text{Tr}(\mathbf{a}_c^\dagger \mathbf{a}_c \boldsymbol{\xi})) \right) = -\mathbb{E} \left(\text{Tr} \left(\left(\sum_c \sqrt{\kappa_c} \mathbf{a}_c \right)^\dagger \left(\sum_c \sqrt{\kappa_c} \mathbf{a}_c \right) \boldsymbol{\xi} \right) \right) \leq 0.$$

$\sum_c \text{Tr}(\mathbf{a}_c^\dagger \mathbf{a}_c \boldsymbol{\xi})$ is not a strict Lyapunov function towards $\mathcal{I}_S$. We guess that exponential convergence towards $\mathcal{I}_S$ is equivalent to assume that, for $u \equiv 0$ and for each s , the linear system made of the n_C equations (8) indexed by c converges exponentially to 0. It will be interesting to characterize the asymptotic regime of the stochastic master equation (9) from the asymptotic regimes of the deterministic ordinary differential equations (8).

The state space input/output description when the initial system/cavity is $\rho(0) = \rho_S^0 \otimes (\otimes_c |\alpha_c^0\rangle\langle \alpha_c^0|)$ ($\alpha_c^0 \in \mathbb{C}$ arbitrary) is given by $\boldsymbol{\xi}_S$ with initial condition ρ_S^0 and by $\alpha_{s,c}$ with initial condition α_c^0 , independent of s . The partial trace versus the system gives the non-Markovian evolution of $\rho_C = \text{Tr}_{\text{system}}(\rho)$ via the following cavity output map

$$\rho_C(t) = \sum_s \text{Tr}(\mathbf{P}_s \boldsymbol{\xi}_S(t)) (\otimes_c |\alpha_{s,c}(t)\rangle\langle \alpha_{s,c}(t)|)$$

The partial trace versus the cavity gives the non-Markovian evolution of $\rho_S = \text{Tr}_{\text{cavity}}(\rho)$ via the following output map:

$$\rho_S(t) = \sum_{s,s'} c_{s's}(t) \mathbf{P}_s \boldsymbol{\xi}_S(t) \mathbf{P}_{s'}$$

where $c_{s's} = \prod_c \langle \alpha_{s',c} | \alpha_{s,c} \rangle$. With $C = (c_{s',s})_{1 \leq s', s \leq n_C}$ the Gram matrix of the n_C states $\otimes_c |\alpha_{s,c}\rangle$ and with its square root $\sqrt{C} = (r_{s',s})_{1 \leq s', s \leq n_C}$, we recover a usual Kraus map formulation

$$\rho_S(t) = \sum_s \mathbf{K}_s(t) \boldsymbol{\xi}_S(t) \mathbf{K}_s^\dagger(t) \quad \text{where} \quad \mathbf{K}_s(t) = \sum_{s'} r_{s,s'}(t) \mathbf{P}_{s'}.$$

5 Concluding remark

It will be interesting to investigate the qubit/cavity system governed by (1) but with a finite life time T_1 for the qubit:

$$d\rho = \left(-i [\sqrt{\kappa} u \mathbf{a}^\dagger + \sqrt{\kappa} u^* \mathbf{a} + \chi \boldsymbol{\sigma}_z \mathbf{a}^\dagger \mathbf{a}, \rho] + \kappa \mathcal{D}_{\mathbf{a}}(\rho) + \frac{1}{T_1} \mathcal{D}_{|g\rangle\langle e|}(\rho) \right) dt + \sqrt{\eta \kappa} \mathcal{M}_{\mathbf{a}}(\rho) dw$$

When $\kappa \gg \frac{1}{T_1}$, an approximate formulation might be possible via perturbation techniques. It might be based on a usual diffusive SME for a fictitious qubit of state $\boldsymbol{\xi}_Q$, on deterministic ordinary differential equations and on Kraus maps to approximate the evolution of the density operators attached to the cavity and to the qubit.

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